

CLASSIFICATION	VEHICLE DESCRIPTION	TOLL WITH TRANSPONDER	TOLL WITH REGISTERED PLATE	TOLL WITH UNREGISTERED PLATE
Passenger Vehicle	2-axle up to 7 1/2 feet in height	\$2	\$3	\$4
Medium Vehicle	2-axle more than 7 1/2 feet in height	\$5	\$6	\$7
	All 3-axle			
	All 4-axle			
Large Vehicle	5-axle or more	\$10	\$11	\$12

Fig. 5: A sophisticated AVI system enables counting the number of axles (Credit: <https://riverlink.com>)

dual-tyre vehicles. Light-curtain laser profilers are used to record the shape of the vehicle—this helps distinguish between, say, trucks and trailers.

Transaction processing system. Transaction processing is used for maintaining customer accounts, posting toll transactions and customer payments to accounts, and handling customer inquiries. Customer accounts may be postpaid or prepaid. Postpaid accounts are where toll transactions are periodically billed to the customer. Whereas, prepaid ones are those where the customer funds a balance in his/her account, which is then depleted as toll transactions occur.

A prepaid system is more commonly used, as low amounts of most tolls makes follow up of uncollected debts expensive. A postpaid account usually insists on a security deposit, effectively rendering it prepaid. Transaction processing function is also referred to as customer service. In many ways, transaction processing function is like conventional banking—at times, toll agencies contract out transaction processing to a bank.

Violation enforcement system (VES). An unmanned toll gate represents an appealing target for toll evasion. VES is used for reducing unpaid tolls. Several methods are in use to detect and discourage toll violators. Enforcement patrols are often deployed effectively at toll gates.

However, this can be quite expensive. A physical barrier, such as a gate arm, can be employed to ensure that all vehicles passing through the toll booth pay the required toll. Violators

get identified immediately, as the barrier does not allow them to proceed. However, barriers also require authorised customers, which are the mainstream of vehicles passing through, to slow down to a near-stop at the toll gate, adversely affecting many of the benefits of electronic tolling.

Automatic number plate recognition can also be used for violation enforcement. A small number of images of violators collected makes manual review quite practical. However, there may be legal hurdles to permit this type of enforcement, as the number plate identifies only the vehicle, not its operator, and traffic enforcement regulations may require identifying the operator to take legal action.

Electronic toll collection system using RFID

In most cases, two methods of collecting toll are generally used. First is the common manual method, where the vehicle passing through the toll plaza stops, and one person collects money and issues a receipt.

The other method is the cashless RFID-based system for automatic collection of toll. In this method, vehicles passing through need not stop and interact with a person to pay the toll. The toll is automatically deducted from their accounts.

Separate electronic toll collection lanes are set up, equipped with scanners that continuously send out RF electronic pulses. These pulses identify

each vehicle that is travelling through the toll plaza with an electronic transponder (RFID tag in the form of a smart card), having all required information/data regarding the user's account, fitted on the windshield of the vehicle.

RF pulses are reflected on hitting a tag and are received by the antenna of the scanner. These reflected pulses from the tag contain information about the user's vehicle number, account, balance and so on. After encrypting the pulses, the toll plaza sends these to a central location, where computers connected to a central server process the information received to detect the unique identification number of the vehicle and the balance available in the user's account. Then, the required toll amount is deducted from the user's account.

The RFID tag-based system carries out the functions of automatic toll collection and detection of vehicle theft, any violation of traffic signals by vehicles, including over-speeding vehicles. When registering a vehicle, a smart card with an RFID-enabled tag with a unique ID is provided to the owner, to be used exclusively with this vehicle. The service provider creates an account for that smart card and maintains transaction history in its database. The owner of the vehicle needs to maintain a minimum balance in the account. Every time the registered vehicle approaches the toll booth, sensors at the toll gate detect the presence of the vehicle and activate the system to read the RFID smart card and calculate the toll fee to be paid by the vehicle. If the balance is available, toll fee is deducted directly; if not, the vehicle is directed to another lane to pay the toll manually. The software updates the details in the centralised database server and triggers a mechanism to generate the bill, which is sent to the user as text message.

If a vehicle owner registers a stolen vehicle complaint on the website using its registration ID and unique RFID tag number,

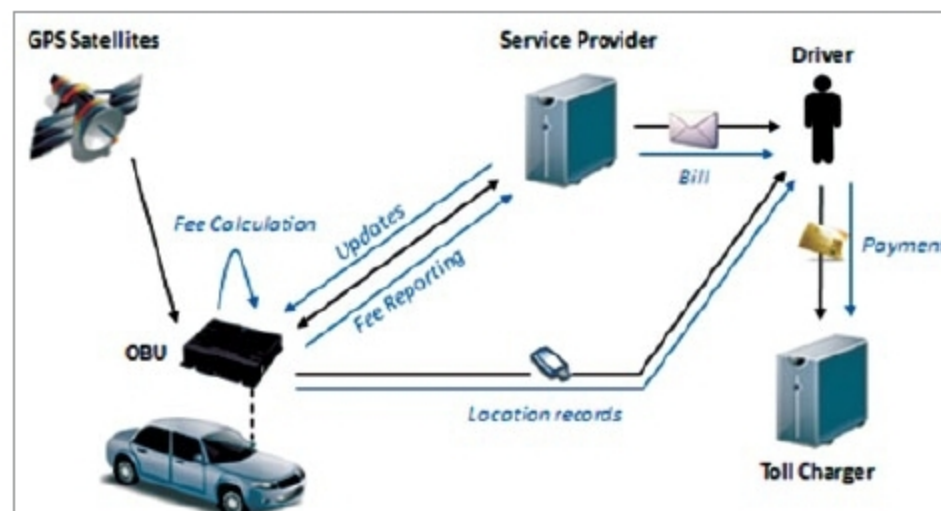


Fig. 6: Typical GPS-based toll collection system (Credit: <https://cosic.esat.kuleuven.be>)